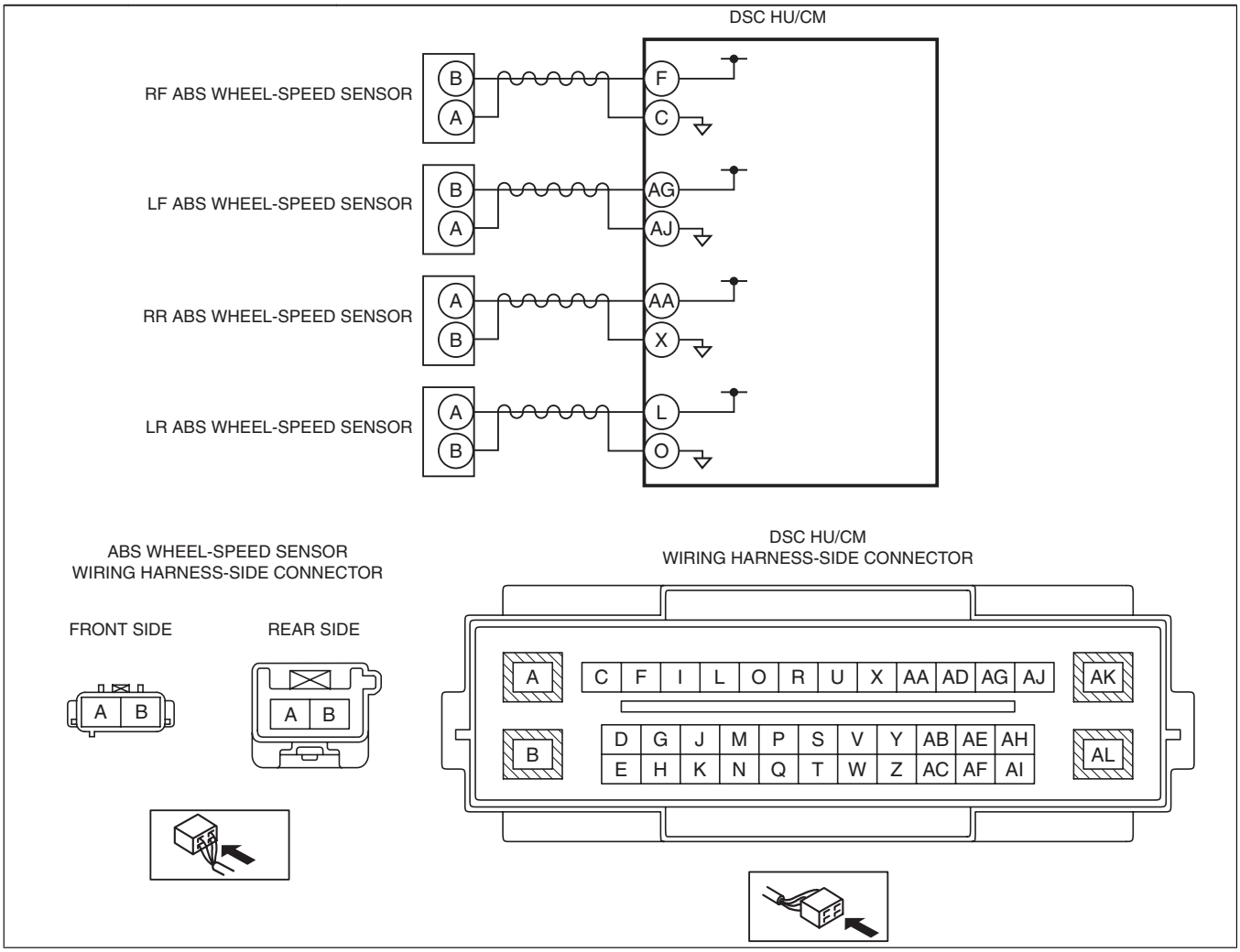
## DTC C0030:07/C0031:07/C0031:29/C0031:2F/C0031:64/C0033:07/C0034:07/C0034:29/C0034:2F/C0034:64/C0036:07/C0037:07/C0037:29/C0037:2F/C0037:64/C0039:07/C003A:07/C003A:29/C003A:2F/C003A:64 [DSC HU/CM]

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| | | |
|---|---|---|
| DTC | C0030:07 | LF ABS sensor rotor |
| | C0031:07 | LF ABS wheel-speed sensor/ABS sensor rotor |
| | C0031:29 | |
| | C0031:2F | |
| | C0031:64 | |
| | C0033:07 | RF ABS sensor rotor |
| | C0034:07 | RF ABS wheel-speed sensor/ABS sensor rotor |
| | C0034:29 | |
| | C0034:2F | |
| | C0034:64 | |
| | C0036:07 | LR ABS sensor rotor |
| | C0037:07 | LR ABS wheel-speed sensor/ABS sensor rotor |
| | C0037:29 | |
| | C0037:2F | |
| | C0037:64 | |
| | C0039:07 | RR ABS sensor rotor |
| | C003A:07 | RR ABS wheel-speed sensor/ABS sensor rotor |
| | C003A:29 | |
| | C003A:2F | |
| | C003A:64 | |
| DETECTION CONDITION | | • C0030:07, C0033:07, C0036:07, C0039:07<br>— Periodic abnormality is detected in the signal wave pattern from the ABS wheel-speed sensors<br>• C0031:07, C0034:07, C0037:07, C003A:07<br>— A large, sudden change in the wheel speed-signal is detected<br>• C0031:29, C0034:29, C0037:29, C003A:29<br>— While the vehicle is traveling at a speed of 10 km/h {6.2 mph} or more, no signal in any of the four wheels or an extremely low vehicle speed signal is detected<br>• C0031:2F, C0034:2F, C0037:2F, C003A:2F<br>— The wheel speed or acceleration speed in any of the 4 wheels is not within the specification<br>— ABS control continues to operate for 28 s or more<br>• C0031:64, C0034:64, C0037:64, C003A:64<br>— While the vehicle is traveling at a speed of 20 km/h {12 mph} or more, an extremely high vehicle speed signal in any of the four wheels is detected |
| FAIL-SAFE FUNCTION | | Refer to DTC TABLE. (See DTC TABLE [DSC HU/CM].) |
| POSSIBLE CAUSE | | • Short to GND in wiring harness between ABS wheel-speed sensor and DSC HU/CM<br>• Excessive clearance between the ABS wheel-speed sensor and sensor rotor<br>• ABS wheel-speed sensor malfunction<br>• ABS sensor rotor malfunction (missing ABS sensor rotor teeth due to foreign material obstruction)<br>• ABS sensor rotor installation malfunction (If the ABS sensor rotor is installed at an angle, it may cause output of abnormal wave pattern at high speeds)<br>• Continuous ABS operation |




## Diagnostic Procedure

| Step | Inspection | | Action |
|---|---|---|---|
| 1 | **INSPECT ABS WHEEL-SPEED SENSOR OUTPUT**<br>• Using the M-MDS, perform the PID inspection as followings:<br>(See PID/DATA MONITOR INSPECTION [DSC HU/CM].)<br>— WSPD_SEN_LF<br>— WSPD_SEN_LR<br>— WSPD_SEN_RF<br>— WSPD_SEN_RR<br>• Drive the vehicle.<br>• Are the vehicle speeds approximately the same? | Yes | Go to Step3. |
| | | No | Go to the next step. |

| Step | Inspection | | Action |
|---|---|---|---|
| 2 | **INSPECT ABS WHEEL-SPEED SENSOR SIGNAL CIRCUIT FOR SHORT TO GROUND**<br>• Disconnect the ABS wheel-speed sensor connector.<br>• Inspect for continuity as followings:<br>— RF ABS wheel-speed sensor terminal A (wiring harness-side)—body ground (SKYACTIV-G 1.3, SKYACTIV-G 1.5)<br>— LF ABS wheel-speed sensor terminal A (wiring harness-side)—body ground (SKYACTIV-G 1.3, SKYACTIV-G 1.5)<br>— RR ABS wheel-speed sensor terminal A (wiring harness-side)—body ground (SKYACTIV-G 1.3, SKYACTIV-G 1.5)<br>— LR ABS wheel-speed sensor terminal A (wiring harness-side)—body ground (SKYACTIV-G 1.3, SKYACTIV-G 1.5)<br>• Is there continuity? | Yes | Refer to the wiring diagram and verify whether or not there is a common connector in the followings:<br>• RF ABS wheel-speed sensor terminal A—DSC HU/CM terminal C<br>• LF ABS wheel-speed sensor terminal A—DSC HU/CM terminal AJ<br>• RR ABS wheel-speed sensor terminal B—DSC HU/CM terminal X<br>• LR ABS wheel-speed sensor terminal B—DSC HU/CM terminal O<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>Go to Step5. |
| | | No | Go to the next step. |
| 3 | **INSPECT CLEARANCE between the ABS wheel-speed sensor and sensor rotor**<br>• Measure the clearance between the ABS wheel-speed sensor and the ABS sensor rotor.<br>(See FRONT ABS WHEEL-SPEED SENSOR INSPECTION.)<br>(See REAR ABS WHEEL-SPEED SENSOR INSPECTION.)<br>• Is the clearance normal? | Yes | Go to the next step. |
| | | No | Replace the malfunction part, then go to Step5. |
| 4 | **VISUALLY INSPECT ABS SENSOR ROTOR**<br>• VISUALLY INSPECT the ABS SENSOR ROTOR FOR FOREIGN MATERIAL ADHERING OR IMPROPER INSTALLATION.<br>• Is the ABS SENSOR ROTOR normal? | Yes | Go to the next step. |
| | | No | Replace the ABS sensor rotor, then go to the next step.<br>• Front ABS sensor rotor (front wheel hub)<br>(See WHEEL HUB, STEERING KNUCKLE REMOVAL/INSTALLATION.)<br>• Rear ABS sensor rotor (rear wheel hub component)<br>(See WHEEL HUB COMPONENT REMOVAL/INSTALLATION [REAR BRAKE (DISC)].) |
| 5 | **VERIFY DTC TROUBLESHOOTING COMPLETED**<br>• Using the M-MDS, clear the DTC from the DSC HU/CM.<br>(See CLEARING DTC [DSC HU/CM].)<br>• Start the engine and drive the vehicle at 10 km/h {6.2 mph} or more.<br>• Using the M-MDS, perform the DSC HU/CM DTC inspection.<br>(See DTC INSPECTION [DSC HU/CM].)<br>• Is the same DTC present? | Yes | Repeat the inspection from Step1.<br>If the malfunction recurs, replace the DSC HU/CM, then go to the next step.<br>(See DSC HU/CM REMOVAL/INSTALLATION [L.H.D.].)<br>(See DSC HU/CM REMOVAL/INSTALLATION [R.H.D.].) |
| | | No | Go to the next step. |
| 6 | **VERIFY NO DTC IS PRESENT**<br>• Are any DTCs present? | Yes | Go to applicable DTC inspection.<br>(See DTC TABLE [DSC HU/CM].) |
| | | No | DTC troubleshooting completed. |